Alexander Daniel Rios

Data Scientist · Machine Learning Engineer
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PROFESSIONAL PROFILE

Data Scientist & ML Engineer with over 4 years of experience in high-impact hands-on projects. Specialized in the full ML model lifecycle — from conception to production — using MLOps architectures (MLflow, GCP, Docker). Expert in applying Deep Learning for NLP and Computer Vision to solve complex real-world problems.

RELEVANT PROJECTS

RAG Research Assistant – AI for Scientific Literature *Sep 2025 – Oct 2025*

- Built a RAG system over **120k+ documents**, optimizing retrieval in **Qdrant** with **Hit Rate 0.91** and **MRR 0.78**.
- Automated ingestion & embeddings pipeline with **FastEmbed and Prefect**, ensuring corpus reproducibility.
- Integrated **Gemini 2.5** generation with real-time monitoring via **BigQuery and Grafana**.

Tech: Python, Qdrant, Gemini, Prefect, BigQuery, Grafana, Streamlit, Docker, FastAPI **Links:** Repository

MLOps Pipeline – Cardiovascular Disease Prediction *Jul 2025 – Aug 2025*

- Built an E2E clinical risk prediction pipeline achieving **accuracy 0.89** with an optimized KNN model.
- Deployed automatically to **GCP Cloud Run** via Docker and Terraform, with Slack *drift* alerts.
- Orchestrated training & monitoring with **MLflow and Prefect** for production model governance.

Tech: MLflow, Prefect, FastAPI, Docker, Terraform, GCP, Evidently AI **Links:** Repository | API

Computer Vision – Blood Cell Cancer Diagnosis *Dec 2024 – Jan 2025*

- Designed a **UNET** architecture for leukemia segmentation & classification: **ROC AUC 0.99**, **BinaryIoU 0.94**.
- Developed a real-time inference tool for hematology, reducing manual microscopic sample analysis time.
- Deployed with **Kubernetes** and Flask for high-availability interactive web access.

Tech: TensorFlow, Keras, Streamlit, Flask, Docker, Kubernetes **Links:** Repository | App

NLP – Disaster Tweet Classification *Jul 2024 – Aug 2024*

- Built a hybrid **BERT + LSTM** model for real-time crisis detection, outperforming baselines by **12%**.
- Processed and vectorized **10k+ tweets** with spaCy, optimizing text cleaning for noisy data environments.
- Implemented a Streamlit dashboard for bulk visualization and classification of critical alerts.

Tech: Transformers (BERT), TensorFlow, spaCy, scikit-learn, Streamlit **Links:** Repository | App

EDUCATION

B.Sc. Electronic Engineering, *UTN — National Technological University* *Apr 2014 – Present (Expected 2026)*
Buenos Aires, Argentina

University Technician in Electronics, *UTN — National Technological University* *Apr 2014 – Mar 2018*
Buenos Aires, Argentina

ADDITIONAL TRAINING

Machine Learning Zoomcamp, *DataTalks.Club* *Sep 2024 – Jan 2025*
End-to-end ML Engineering: modeling with Scikit-Learn/TensorFlow, deployment with Docker, Kubernetes, and serverless.

Machine Learning Operations Zoomcamp, *DataTalks.Club* *May 2025 – Aug 2025*
MLOps: experiment tracking (MLflow), pipeline orchestration, monitoring (Evidently AI), IaC on GCP with Terraform.

AI Agents, *Hugging Face* *May 2025 – Jun 2025*
Design and implementation of AI agents using LangGraph and LlamaIndex.

ARTICLES & PUBLICATIONS

Computer Vision for Brain Tumor Diagnosis — *LinkedIn* *2023*
How to Build a Blood Cell Classifier for Cancer Prediction — *DataTalks.Club Blog* *2025*
Natural Language Processing with spaCy, TensorFlow and BERT Model Architecture — *Notion* *2025*

LANGUAGES

Spanish	Native
English	B1 Intermediate — EF SET Certificate